

Master Thesis — Progress Report

Spain's MTU15 reform: empirical assessment in progress

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May 2026

Outline of the talk

- **1. Reminder.** Objective, research question, context.
- **2. What I have done.** Data, methods, headline findings.
- **3. State of the work.** Claim portfolio, theoretical model.
- **4. Difficulties encountered.** Identification, confounds, surprises.
- **5. What is left.** Empirical, theoretical, writing.

Format

18–20 min, following advisor's guidance: a progress report rather than a results talk.

Objective and research question

One-sentence aim

Empirically assess the impact of Spain's 2024–2025 **electricity-market clock reform** (the *MTU15 reform sequence*) and extend the theoretical framework to better understand firms' strategic behaviour under different market-time-unit (MTU) configurations.

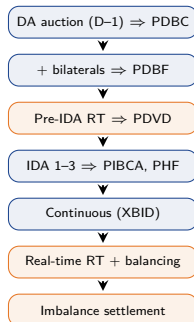
Research question

- How did the reform affect **prices, quantities, liquidity, imbalances**, and **firm strategic behaviour**?
- Are the welfare and competition implications consistent with the theory?
- Which transitional configurations are most distortionary, and why?

Big picture: the Spanish electricity market

For each delivery hour, the wholesale market clears **sequentially** across D-1 and D:

- **[OMIE]** Day-ahead auction (DA) $\Rightarrow$ PDBC
- + bilateral contracts $\Rightarrow$ PDBF
- **[REE]** pre-IDA technical restrictions $\Rightarrow$ PDVD
- **[OMIE]** Intraday auctions IDA-1, IDA-2 (D-1) and IDA-3 (D)
- **[OMIE]** Continuous market (XBID/SIDC), opens after IDA-2
- **[REE]** Real-time technical restrictions and balancing
- **Imbalance settlement:** ex-post, dual-pricing rule (EBGL Art. 52)



Each clearing produces a *firm program* that the next can adjust.

The 2024–2025 reform sequence

- Traditionally the market time unit (MTU) was **60 min** everywhere.
- The clock rewiring happened in **three stages**:
 - **Dec 2024 — ISP15**: imbalance settlement becomes 15-min.
 - **Mar 2025 — ID15**: intraday auctions and continuous become 15-min.
 - **Oct 2025 — DA15**: day-ahead becomes 15-min.
- Two *transitional regimes* where settlement and trading granularities differ:
 - **ISP15-win** (Dec 2024 – Mar 2025): DA60/ID60 + ISP15
 - **DA60/ID15** (Mar 2025 – Oct 2025): DA60 + ID15 + ISP15

Period	ISP	DA	ID
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	60	15
Oct25 onward	15	15	15

Minutes; 60 = hourly, 15 = quarter-hourly.

Plus an *operational regime overlay*: the **post-blackout *operación reforzada*** (Apr 28, 2025 onward) keeps tightening REE security criteria via PO-3.2 RRTT.

Data pipeline

- **OMIE** (Iberian market operator): full short-term data, 2018-01 → 2026-02.
 - Day-ahead, intraday auctions, continuous, prices, programs, bid curves.
 - All families parsed; 1 parser per family, idempotent pipelines.
 - **Recently closed gap**: PDBF (PDBC + bilateral contracts) ingested → **166M rows**, opens a new strategic-channel dimension.
- **ESIOS / REE**: settlement detail, RT prices, aFRR offers, unit outages.
- **ENTSO-E**: pan-European balancing, generation by type, A75 / A73 / A86 / A87.
- **AEMET / weather**: temperature, wind, solar (controls).

Discipline

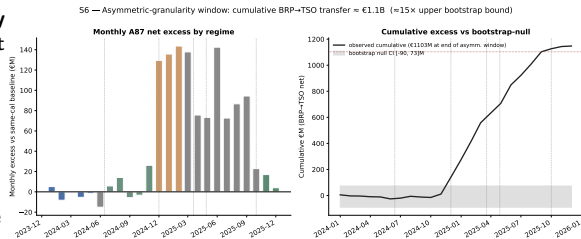
Repo-relative paths; raw → processed → derived layers separated; analytical outputs (regressions, figures) under `results/` and `figures/` (never under `data/`).

Empirical strategy

- **No clean external counterfactual:** most comparable European markets adopted MTU15 already and differ structurally.
- Reduced-form before/after with rich controls:
 - Fundamentals: load, wind, solar (actual + forecast errors), temperature, fuel/CO₂.
 - Calendar structure: cal-month FE, day-of-week, hour-of-day, year FE (where appropriate).
 - Reform-date indicators (Dec 2024 / Mar 2025 / Oct 2025) and *reforzada* indicator.
- **Same-calendar-month robustness:** mandatory for any cross-regime claim (Spanish electricity is highly seasonal).
- **Cluster-robust SE:** by date or week-of-sample; Cameron–Miller $G \geq 30$ rule respected.
- **Cross-country placebo:** France DA prices flat across Spanish reform dates.
- **Structural-time-series counterfactual** (Brodersen et al. 2015) on key outcomes *ongoing*.

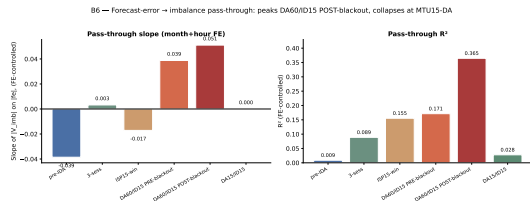
Headline finding 1 — system-layer settlement transfer

- During the **10-month asymmetric-granularity window** (Dec 2024–Sep 2025), DA cleared at 60-min while ID and imbalance settled at 15-min.
- Net BRP→TSO settlement transfer (A87): **+€1,094.9M** cumulative, vs same-calendar pre-IDA baseline.
- Bootstrap null CI [-90, +73] M $\Rightarrow \approx 15\times$ the upper bound.
- **Collapses to €7.4M/mo at MTU15-DA** ($12\times$ drop) once the clock asymmetry is closed.
- This is a *regulatory redistribution*, not a deadweight loss.



Headline finding 2 — the mechanism (forecast-error pass-through)

- **B6:** pooled regression
 $|V_{imb,t}| \sim |fe_t| + R \times |fe_t| + FE$, daily grain,
cluster SE by year-month.
- Net per-regime slope β :
 - pre-IDA ~ 0
 - 3-sess ~ 0
 - ISP15-win ≈ -0.03
 - **DA60/ID15 $\approx +0.09$** — uniquely
elevated
 - DA15/ID15 ~ 0
- **Robust to year FE.** The DA60/ID15 elevation is
the structural break.
- Reframed (recently): the cleaner statement is the
slope, not the R^2 jump.



Headline finding 3 — firm-level market power

- **F7** — Ciarreta–Espinosa synthetic-firm replication.
 - Substitute each Big-4 plant's offer with a same-tech, same-capacity Fringe plant; re-clear the auction.
 - **Iberdrola alone** accounts for ~€820M DA cleared-price-difference rent post-MTU15-IDA, ~98% of joint Big-4 transfer.
 - Hydro 64% / CCGT 36%; survives 5 caveat tests (BRP migration, hydro-strict matching, blackout split, complex-offer exclusion, pre-reform 0-padded).
 - **Regime-invariant:** the rent does not arise from MTU15 reform; it is a persistent structural feature of the Spanish DA market.
- Behavioural complement (B9, just re-validated):**
- Big-4 voluntary IDA repositioning (q_2) shows progressive collapse across regimes; recovers partially at DA15/ID15.
 - Robust to (i) PDBF q_1 controls, (ii) year FE, (iii) canonical Fabra-Imelda spec with forecast-error controls + week-of-sample cluster: regime coefficients vary $< 0.1\%$.

Claim portfolio (CLAIMS_LEDGER.md)

- **68 active rows** tracked formally. Status (post 2026-05-02 attack pass):
 - **41 alive** — passed all documented robustness checks.
 - 7 wounded — survive in narrowed form; cited with caveat.
 - 20 dead — retracted or contradicted; kept as audit trail.
- Each cited number is regression-backed at proper grain (firm-day or finer), date- or week-clustered SE, and survives multi-spec robustness.

Three policy levers identified:

1. **Clock-symmetry** (lever 1): *implemented* at MTU15-DA, closes the asymmetric-window transfer.
2. **Pigouvian incidence** (lever 2): *open*. Wind + retailers consistently bear 60–65% of imbalance € regardless of clock symmetry.
3. **Operational regime overlay** (lever 3): *operación reforzada* — the post-blackout missing-market patch. New PO-7.4 zonal reactive market (Jun 2025) is the first market-design response.

Theoretical model — where it stands

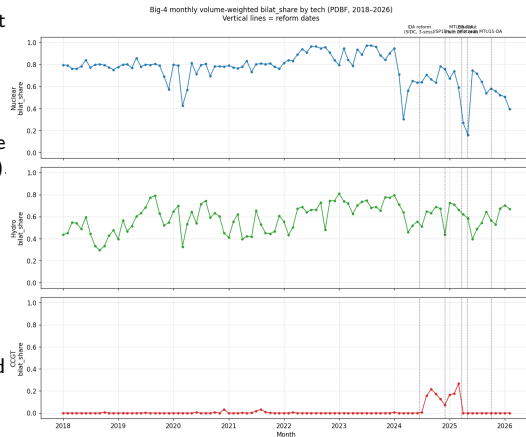
- Started from my Energy Economics research proposal: an extension of **Ito–Reguant (2016)** sequential-market model with an explicit role for the MTU mismatch.
- Predicted (i) DA15 is the key reform that smooths imbalances; (ii) dispersion risk concentrates in DA60/ID15; (iii) finer granularity creates winners and losers.
- Empirically, all three predictions are **borne out qualitatively**; the magnitudes are larger than the theory suggested.
- **Pivot**: the project focuses on *very solid empirics* + a *simplified IO theory model* to illustrate the mechanism + a light structural exercise (FOC-based, in the spirit of HP-sophistication or Allaz–Vila).
- **Not** a fully estimated structural auction model (Hortaçsu–Puller / Wolak): out of scope for the time available.

Difficulty 1 — identification under multiple confounds

- Each reform date has a competing event:
 - **Dec 2024** (ISP15) — concurrent with European gas-price normalisation.
 - **Mar 2025** (ID15) — concurrent with **CNMC Rule 28.8 elimination** (changed bilateral-contract rules).
 - **Apr 28, 2025** blackout $\Rightarrow$ *operación reforzada* (regulatory regime overlay continuing through the talk).
 - **Oct 2025** (DA15) — partial overlap with reforzada.
- **Identification spine adopted:**
 - *PDBC vs PDVD chain placement* — reforzada acts *after* OMIE clears PDBC, so PDBC-side outcomes face it only through expectations.
 - *Reforzada-held-constant pairings*: ISP15-win $\rightarrow$ DA60/ID15-PRE-blackout (both pre-reforzada); DA60/ID15-POST $\rightarrow$ DA15/ID15 (both under reforzada).
 - Same-calendar-month robustness, cross-country placebo (B7), bootstrap CIs.
- Honest record of **14 dead identification attempts** (X1–X14): TWFE-DiD, wind-IV, profile-matching, etc. — each rejected after placebo / parallel-trends / randomization tests.

Difficulty 2 — a structural break we cannot yet explain

- Big-4 **nuclear bilateral share** (PDBF) was stable at 80–95% throughout 2018–2023.
- Drops sharply in **Feb–Mar 2024**, three months *before* any MTU15 reform.
- Hydro shows the same Q1-2024 break (auction-side growth); CCGT is unaffected (always 0% bilateral).
- Concurrent with the **end of the Iberian gas-cap (TPP) on 2023-12-31** and Spain's first major renewable-saturation event (negative-price share emerges in April 2024).
- Regression: $\beta(\text{neg-price-share, exogenous proxy}) = -3.05$ on nuclear bilat_share ($p \approx 10^{-179}$, $R^2=0.13$) — the magnitude matches the observed drop. Hydro: $\beta \approx 0$.



Mechanism unidentified — likely renewable-saturation onset interacting with PPA structures, but causal identification still open. Surfaced this week and reframes how I read 2024 pre-reform data.

Difficulty 3 — *operación reforzada* as missing-market patch

- Post 2025-04-28 blackout, REE imposes a continuous **reinforced-operation regime**:
 - PO-3.2 RRTT redispatch (forces ~20–30 CCGTs + nuclear daily).
 - PO-1.5 enlarged secondary band; new PO-7.4 zonal reactive market (Jun 2025).
 - Direct cost ~€666M (May 2025 – Mar 2026); RRTT cost €3.77B in 2025 (+49% YoY).
- Affects **all post-blackout regressions**, but in a structurally identifiable way:
 - Where the outcome is computed at PDBC (e.g., F7), *reforzada* enters only via expectations.
 - Where it is computed at PDVD or PHF, *reforzada* enters mechanically.
- **Cleanest empirical separator**: the S6 transfer collapses at MTU15-DA *while reforzada is still active* ⇒ separates lever-1 (granularity) from lever-3 (operational overlay).

Other surprises (briefly)

- **F1/F2/F3 killed**: implied Cournot-FOC Lerner does not match realised marginal-bid markup (Hortaçsu–Puller-style sophistication test). The $+0.318$ / $+0.135$ regime coefficients are formula-mechanical, *not* realised conduct.
- **F5 killed**: the Allaz–Vila commitment slope is a mechanical accounting identity ($Q \approx q_{DA} + \Delta Q_{IDA}$) at every granularity I tried.
- **B11 wounded**: the original Rule-28.8 estimate (-41.7%) was a 12-month-window artefact; under wider-window robustness with placebo tests, the cleaner figure is $\sim -12\%$.
- **F23/F25/F26 wounded or killed**: bilateral nuclear-channel attribution to *operación reforzada* was wrong by 3 months — the structural break is at Q1 2024, not at the IDA reform.

Each item triggered a discipline cycle: ledger update + script header update + research-diary entry + (where relevant) wider-claim reinterpretation.

What is left — empirical

- **Q1 2024 mechanism:** pin down what specifically about the gas-cap expiry / renewable-saturation onset / PPA renegotiations drove the bilateral-channel collapse. Likely a separate chapter discussion.
- **Continued attack pass** on the remaining alive claims (F4, F14–F22, B5, B8) using canonical conventions.
- **Structural time-series counterfactual** (Brodersen et al. 2015) on the headline outcomes: wedge, dispersion, liquidity, balancing.
- **Per-segment imbalance €** verification using the `liqsegme` ESIOs file (parsed but not yet used).
- **Workshop figures:** a couple of figures still in `figures/working/`; need polishing for the thesis.

What is left — theoretical and writing

- **Theoretical model:**

- Refine the IR extension to better motivate the *three-lever* framing (asymmetric clock / Pigouvian incidence / operational overlay).
- Possibly add a small heterogeneous-firm extension to motivate D7 (cross-tech bilateral-channel heterogeneity).
- Light **structural exercise**: HP-sophistication-style FOC test on Big-4 bid functions post-MTU15-IDA, no full estimation.

- **Thesis writing:**

- Part I (system-layer reform impact) — drafting starts week of May 5.
- Parts II–V (firm market power, cross-market specialisation, post-CNMC conduct, behavioural appendix) — covered by the existing alive findings; framing already drafted in the proposal.
- Deadline mid-June 2026.

- **Risk**: 8-week timeline tight; Part-II depth (F7 + F10 + F13) is the bottleneck.

Thank you.

Questions and feedback welcome.

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